CHM_159
Worksheet_Chapter_3_ Answers

Electron configurations and the Periodic Table, Periodic trends

General rules for writing electron configurations:

1) **Electrons will reside in the available orbitals of the lowest possible energy.**
2) **Each orbital can accommodate a maximum of two electrons.**
3) **Electrons will not pair in degenerate orbitals if an empty orbital is available.**
4) **Orbitals will fill in the order indicated in the figure.**

electron configurations

1. a. Use the Aufbau principle to obtain the complete ground-state electron configuration for the following elements: B, Cr, Cu

Ans : B $Is^2 2s^2 2p^1$
Cr $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Cu $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

b. Use the Aufbau principle to obtain the condensed ground-state electron configuration for the following elements: Ge, Fe, Zn

Ans : Ge [Ar) $4s^2 3d^{10} 4p^2$
Fe [Ar) $4s^2 3d^6$
Zn [Ar) $4s^2 3d^{10}$

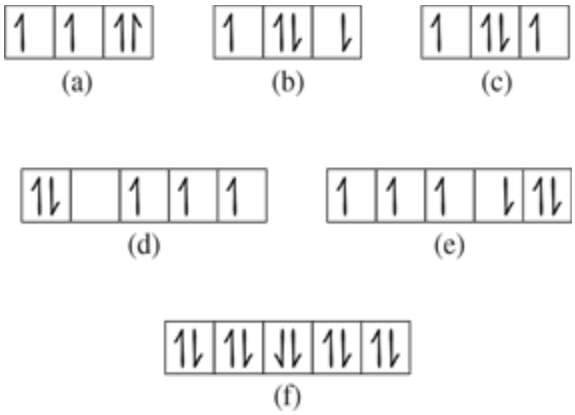
2. The electron configurations described in this chapter all refer to gaseous atoms in their ground states. An atom may absorb a quantum of energy and promote one of its electrons to a higher-energy orbital. When this happens, we say that the atom is in an excited state. The electron configurations of some excited atoms are given. Identify these atoms and write their ground-state configurations:

a. $1s^1 2s^1$ b. $1s^2 2s^2 2p^2 3d^1$ c. $1s^2 2s^2 2p^6 4s^1$ d. $[Ar]4s^1 3d^{10} 4p^4$ e. $[Ne]3s^2 3p^4 3d^1$

Ans :a. He$1s^2$ b. N $1s^2 2s^2 2p^3$ c. Na $1s^2 2s^2 2p^6 3s^1$ d. $[Ar]4s^2 3d^{10} 4p^3$

e. Cl [Ne] $3s^2 3p^5$

3. Portions of orbital diagrams representing the ground-state electron configurations of certain elements are shown here. Which of them violate the Pauli exclusion principle? Which violate Hund's rule?



Ans ; (a) and (f) violate Pauli's Exclusion Principle . (b), (d) and (e) violate Hund's rule

4. Write the ground-state electron configurations of the following ions, which play important roles in biochemical processes in our bodies:
   a. Cl^-,
   b. Fe^{2+},
   c. Cu^{2+},
   d. Zn^{2+}.
   e. Sn^{2+}

Ans : a. [Ar]
b. [Ar] $3d^6$
c. [Ar] $3d^9$
d. [Ar] $3d^{10}$
e. [Kr]$5s^2 4d^{10}$

5. Group the following electron configurations in pairs that would represent elements with similar chemical properties:

a. $1s^2 2s^2 2p^6 3s^2$
b $1s^2 2s^2 2p^3$
c. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
d. $1s^2 2s^2$
e. $1s^2 2s^2 2p^6$
f. $1s^2 2s^2 2p^6 3s^2 3p^3$
Ans : a and d, b and f, c and e

6. A M^{3+} ion derived from a metal in the first transition metal series has five electrons in the 3d subshell. What element might M be?
Ans : Fe

periodic properties

7. a. Which of the following species are isoelectronic with each other:
C, Cl^{-}, Mn^{2+}, B^{-}, Ar, Zn, Fe^{3+}, Ge^{2+}

Ans : (C B^{-}) , (Cl^{-} Ar), (Mn^{2+} Fe^{3+}), (Ge^{2+} Zn)

b. Group the species that are isoelectronic:
Be^{2+}, F^{-}, Fe^{2+}, N^{3-}, He, S^{2-}, Co^{3+}, Ar.

Ans : (Be^{2+} He) , (F^{-} N^{3-}), (Ar S^{2-}), (Fe^{2+} Co^{3+})

8. Arrange the following atoms according to increaasing effective nuclear charge experienced by their valence electrons. Justify your answer.
Al, Si, Mg, S

Ans : S > Si > Al > Mg . All four atoms have the same number of core electrons that contribute to shielding. So the effective nuclear charge will increase with increasing number of protons in the nucleus

9. On the basis of their positions in the periodic table, select the atom with the larger atomic radius in each of the following pairs:

a. Na, Si
b. Ba, Be
c. N, F
d. Br, Cl
e. Ne, Kr

Ans : Na, Ba, N, Br, Kr

10. Arrange the following atoms in order of increasing atomic radius: Na, Al, P, Cl, Mg.
Ans : Cl<P<Al<Mg<Na

11. A certain element has successive ionization energies IE1, IE2, IE3, IE4, IE5 with values, respectively:

1086 kJ/mol, 2353 kJ/mol, 4621 kJ/mol, 6223 kJ/mol, 37831 kJ/mol

This element belongs to which group in the periodic table?

Ans : Group 4A

12. a. Arrange the following in order of increasing first ionization energy:
Na, Cl, Al, S, and Cs.

Ans : Cs< Na< Al< S< Cl

b. Arrange the following in order of increasing first ionization energy:
F, K, P, Ca, and Ne.

Ans : K< Ca< P<F<Ne

13. . a Arrange the elements in each of the following groups in order of increasing electron affinity (more negative value):

a. B, C,N, O, F b. F, Cl, Br, I.

Ans : a. N<B<C<O<F b. I<Br<F<Cl

b.Specify which of the following elements you would expect to have the greatest electron affinity: (more negative value):

He, K, Co, S, Cl.

Ans : Cl

14. Write equations representing the following processes:

a. The electron affinity of S^-

b. The third ionization energy of titanium

c. The electron affinity of Mg^{2+}

d. The ionization energy of O^{2-}

Ans : a. $S^- + e \rightarrow S^{2-}$

b. $Ti^{2+} \rightarrow Ti^{3+} + e$

c. $Mg^{2+} + e \rightarrow Mg^+$

d. $O^{2-} \rightarrow O^- + e$

15. Two atoms have the electron configurations $1s^2 2s^2 2p^6$ and $1s^2 2s^2 2p^6 3s^1$. The first ionization energy of one is 2080 kJ/mol, and that of the other is 496 kJ/mol. Match each ionization energy with one of the given electron configurations. Justify your choice.

Ans : $1s^2 2s^2 2p^6$ Ne 2080 kJ/mol
$1s^2 2s^2 2p^6 3s^1$ Na 496 kJ/mol